SOC 205B Data Analysis in Sociology

Review of Linear Regression

- Basic assumptions
- Hypothesis testing

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Multiple Linear Regression Model

$$y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \ \ldots. + \beta_k X_{ik} + \varepsilon_i$$

y_i → Observed value of the explained variable for individual i
β_0 → Constant or intercept
β_k → Slope parameters (the effect of explanatory variable k on y)
X_{ik} → Observed value of the explanatory variable k for individual i
ε_i → Unobservables affecting y for individual i

Main Assumptions of OLS

- Linearity

Main Assumptions of OLS

Linearity

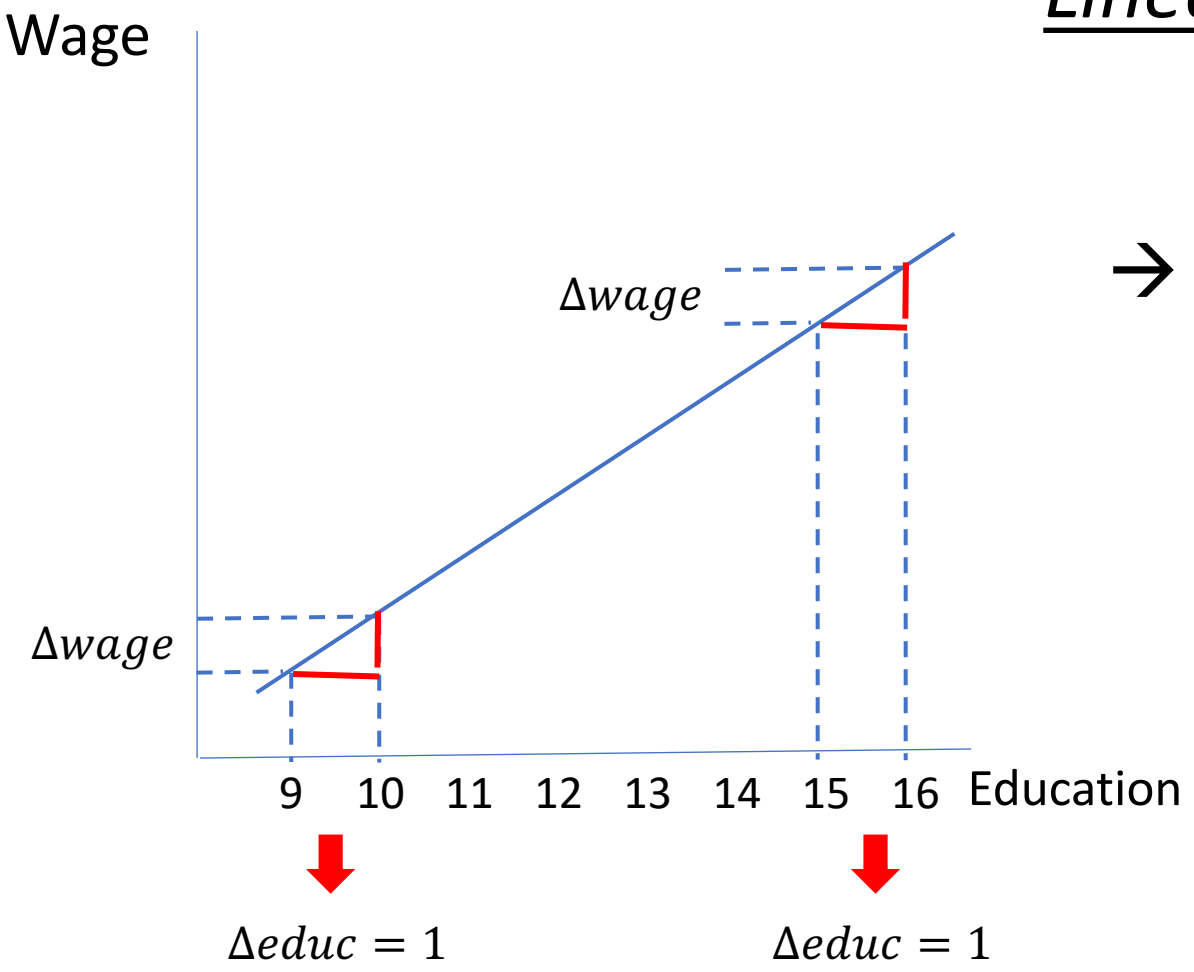
➔The Relationship between dependent and independent variables

linear

➔Dependent variable is assumed to be **continuous** (ranging $-\infty$, $+\infty$)

Main Assumptions of OLS

Linearity



$$wage = \beta_0 + \beta_1 educ + \varepsilon$$

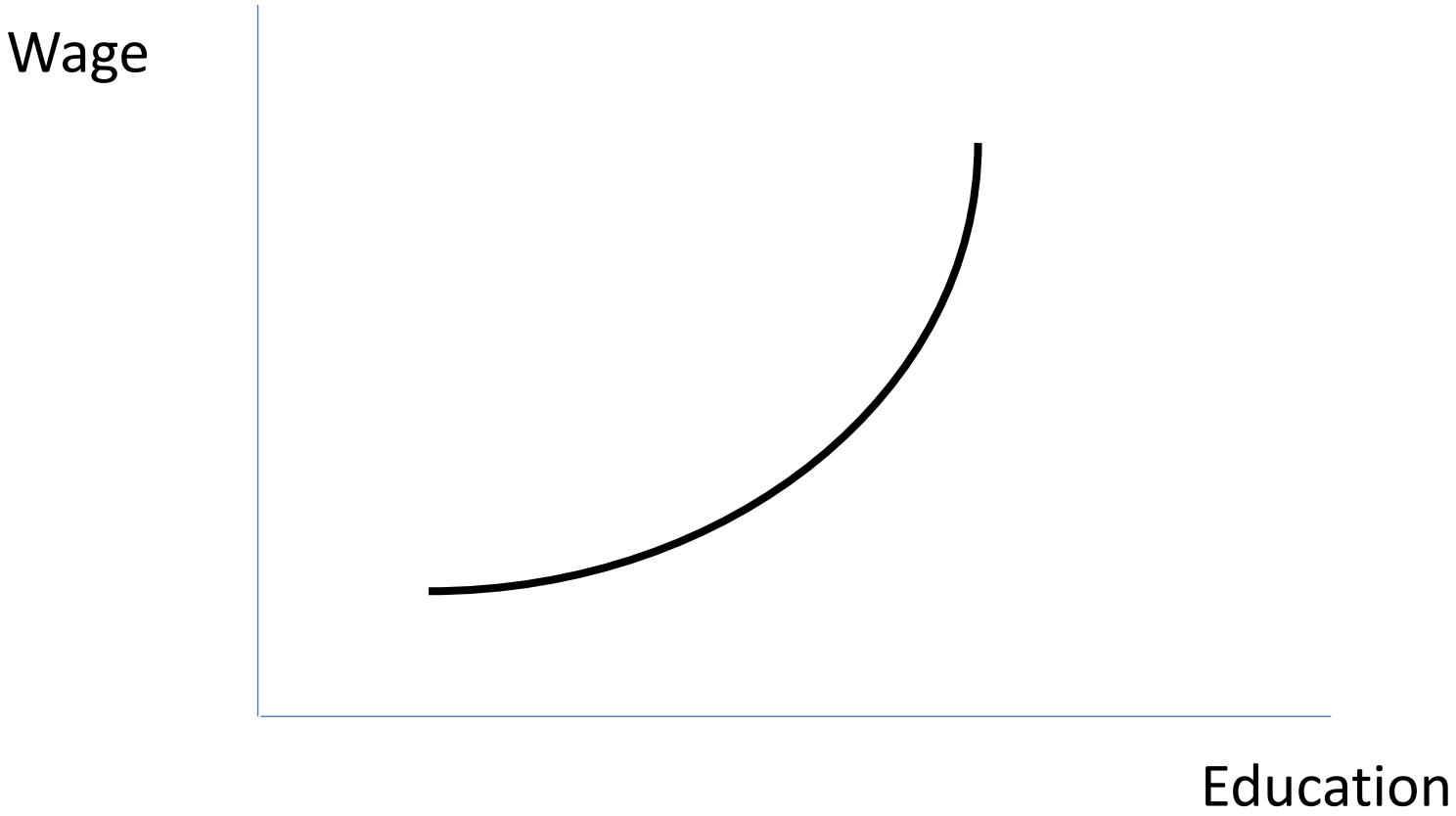
$$\rightarrow \quad \Delta wage = \beta_1 \Delta educ \quad \text{when} \quad \Delta\varepsilon = 0$$

- Is each year of education really worth the same dollar amount no matter how much education one starts with?

Main Assumptions of OLS

Linearity

- However, the relationship may not be linear…



Main Assumptions of OLS

- Linearity
- X’s independent of the errors

Main Assumptions of OLS

<u>X's independent of the errors</u>

$$y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \ \ldots . + \beta_k X_{ik} + \varepsilon_i$$

when $$E\left(\varepsilon_i \middle| X_{i1,} X_{i2,\ldots,} X_{ik}\right) = 0$$

Main Assumptions of OLS

<u>X's independent of the errors</u>

$$wage_i = \beta_0 + \beta_{educ} educ_i + \varepsilon_i$$

Suppose ε_i is "ability".

→ $E(ability|educ = 8) = E(ability|educ = 12) = E(ability|educ = 16)$

- Because people choose education levels partly based on ability, this assumption is almost certainly false.

Main Assumptions of OLS

X's independent of the errors

➔To avoid this problem you would need to include all the relevant explanatory variables in the model.

➔ if X's are not independent of the error, the coefficients will be likely to be overestimated due to omitted variable bias.

Main Assumptions of OLS

<u>X's independent of the errors</u>

$$wage_i = \beta_0 + \beta_{ed} educ_i + \beta_{ab} ability_i + \beta_{ex} exp_i + \beta_g gender_i + \cdots + \varepsilon_i$$

- If we do not include this variable we will overestimating the effect of education on wages.

Main Assumptions of OLS

- Linearity
- X’s independent of the errors
- Independence of observations

Main Assumptions of OLS

Independence of observations

➔Common sources of non-independence include:

- Observations that are close together in time.
- Observations that are close together in space or nested.
- Observations that are somehow related.

Main Assumptions of OLS

Independence of observations

➔ if observations in the sample are not independent standard errors will not be accurate.

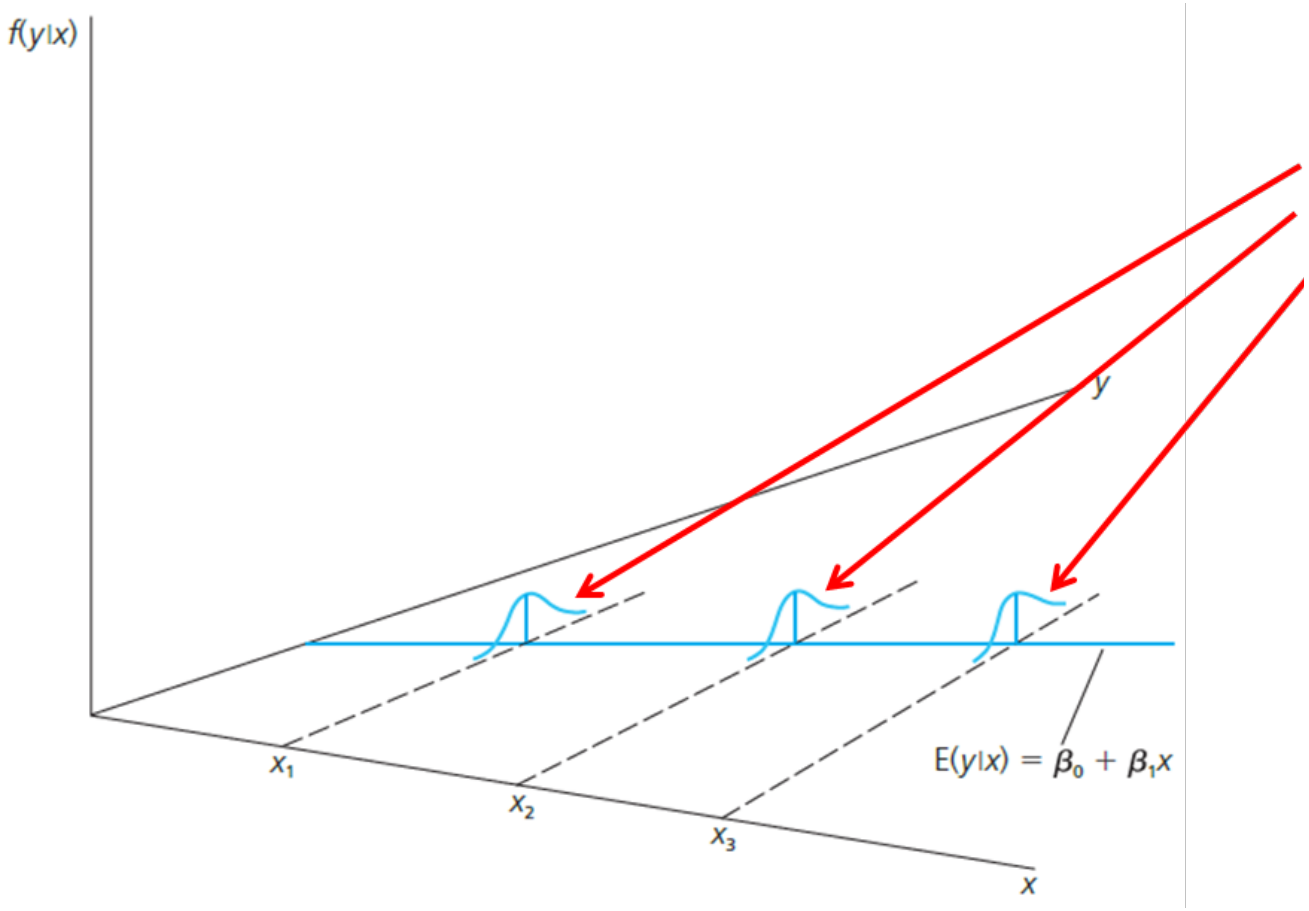
➔Following good sampling techniques (like random sampling) will help to ensure your samples are independent.

Main Assumptions of OLS

- Linearity
- X’s independent of the errors
- Independence of observations
- Constant variance (homoskedasticity of the error term)

Main Assumptions of OLS

- **Constant variance (homoskedasticity)**



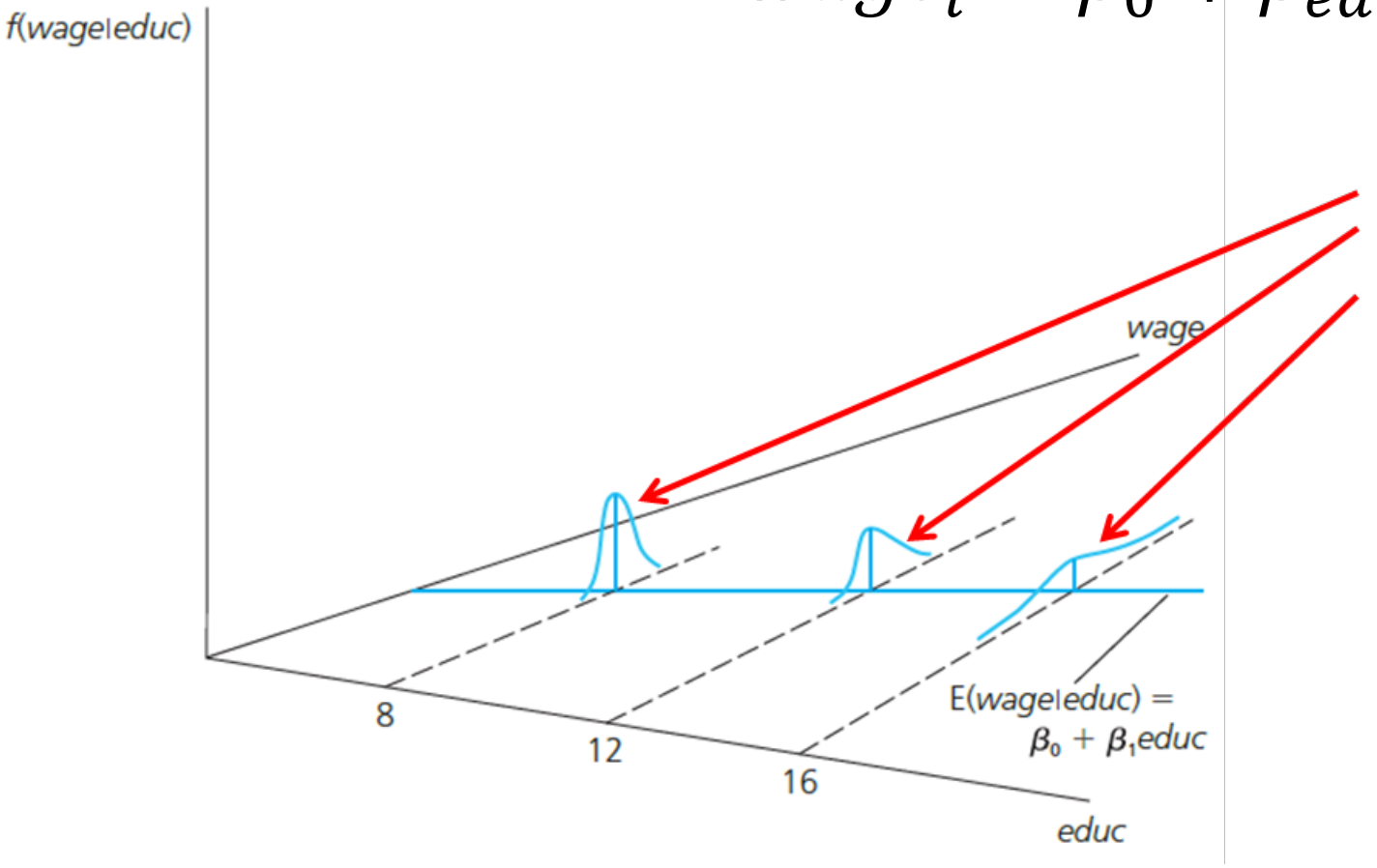
➔The variability of the unobserved factors does not depend on the explanatory variable.

➔Errors are normally distributed when conditioned on x's

Main Assumptions of OLS

- **Heteroskedasticity: e.g. Wage and education**

$$wage_i = \beta_0 + \beta_{educ} educ_i + \varepsilon_i$$



➔The variance of the unobserved determinants of wages increases with the level of education.

Sample data

- Monthly Earnings (in pesos $)
- Years of schooling (in years)
- Household/s head education (in years)

Model 1

- Education has a linear effect on respondent's earnings

$$Earn_i = \beta_0 + \beta_{educ} educ_i + \varepsilon_i$$

Linear regression between earnings and education

Command **Dependent** **Independent**

```
. reg _2earn _2bgedu

      Source |       SS           df       MS      Number of obs   =     2,295
-------------+----------------------------------   F(1, 2293)      =    179.56
       Model |  8.7486e+09         1  8.7486e+09   Prob > F        =    0.0000
    Residual |  1.1172e+11     2,293  48722882.8   R-squared       =    0.0726
-------------+----------------------------------   Adj R-squared   =    0.0722
       Total |  1.2047e+11     2,294  52515329.9   Root MSE        =    6980.2

------------------------------------------------------------------------------
      _2earn |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
     _2bgedu |   446.8497   33.34715    13.40   0.000      381.456    512.2434
       _cons |   625.7047   200.7502     3.12   0.002     232.0337    1019.376
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- **Each additional year of schooling increases monthly earnings in 446 pesos.**

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```

- **Constant: For individuals with zero years of education, predicted monthly earnings are 625 pesos.**

Linear regression between earnings and education

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```

- **R square → The variance in respondent's education explains 7.26% of the variance in earnings.**

- How to test that household head’s education has no effect on earnings, net of respondent’s education (or holding constant respondent’s education)?

How to test that household head's education has no effect on earnings, net of respondent's education?

- Two approaches
  - Estimate (a) $Earn_i = \beta_0 + \beta_1 educ_i + \beta_2 HHeduc_i + \varepsilon_i$
    and Test for β_2=0
    → p-value (β_2) <=0.05 ➔ $\beta_2 \neq 0$

  - F-test for incremental improvement to R^2,
    compare (a) to (b): $Earn_i = \beta_0 + \beta_1 educ_i + \varepsilon_i$

How to test that household head's education has no effect on earnings, net of respondent's education?

- Two approaches
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    → p-value of β_2 <=0.05

Unrestricted model

  - F-test for incremental improvement to R^2,
    compare (a) to (b): $Earn_i = \beta_0 + \beta_1 educ_i + \varepsilon_i$

Restricted model

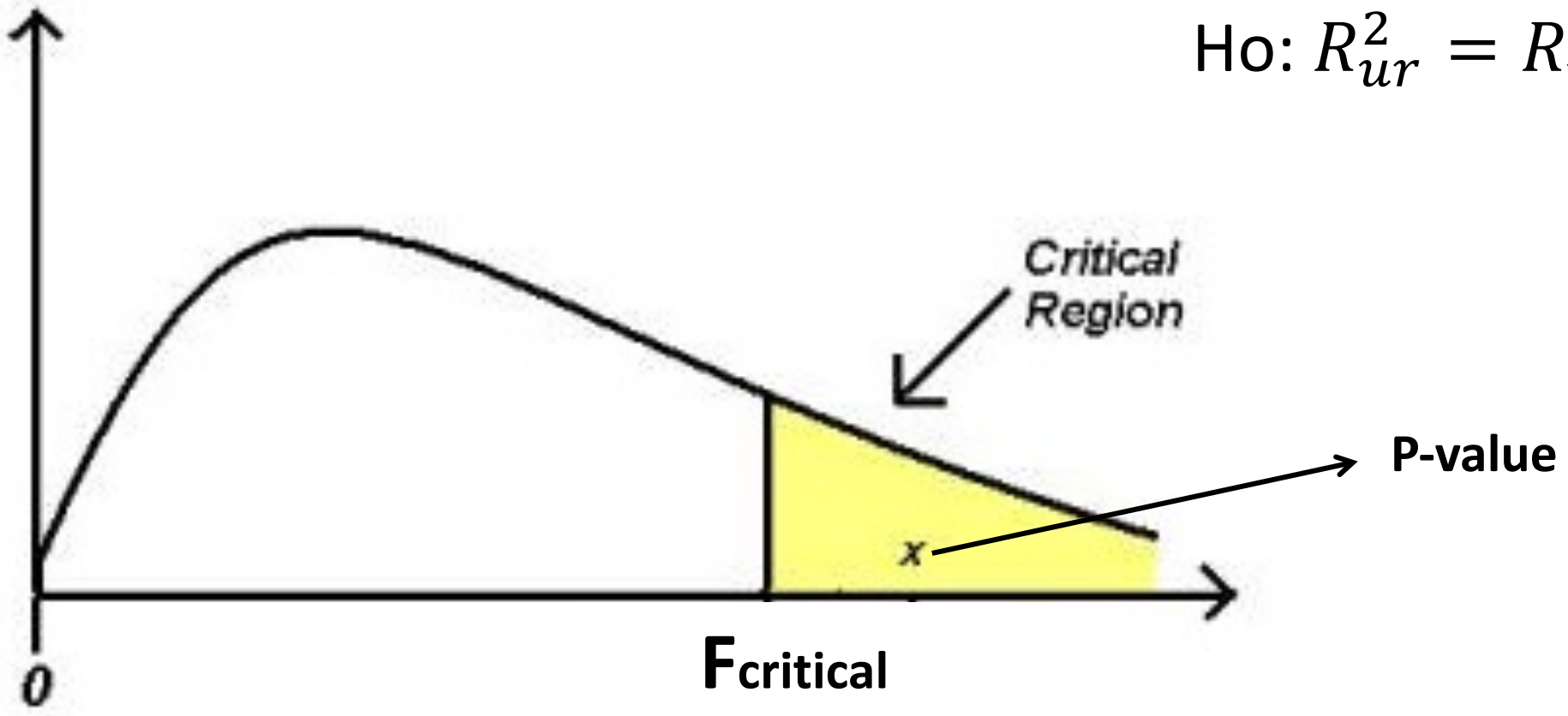
Calculate the **F-statistic** (only appropriate for nested models!!)

$$F_{(k_{ur}-k_r),(N-k_{ur}-1)} = \frac{\frac{R_{ur}^2-R_r^2}{k_{ur}-k_r}}{\frac{1-R_{ur}^2}{N-k_{ur}-1}}$$

k*ur* : Number of independent variables in the unrestricted model.
k*r* : Number of independent variables in the restricted model.
N : Number of observations
R_{ur}^2: R square of the unrestricted model
R_r^2 : R square of the restricted model

If the F statistic > F critical value, you reject the null hypothesis that the two R^2 are the same

Ho: $R^2_{ur} = R^2_r$



(a) $Earn_i = \beta_0 + \beta_1 educ_i + \beta_2 HHeduc_i + \varepsilon_i$

```
. reg _2earn _2bgedu _2jhedu

      Source |       SS           df       MS      Number of obs   =     2,295
-------------+----------------------------------   F(2, 2292)      =     90.10
       Model |  8.7814e+09         2  4.3907e+09   Prob > F        =    0.0000
    Residual |  1.1169e+11     2,292    48729817   R-squared       =    0.0729
-------------+----------------------------------   Adj R-squared   =    0.0721
       Total |  1.2047e+11     2,294  52515329.9   Root MSE        =    6980.7

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      _2earn |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
     _2bgedu |   394.5235   71.94655     5.48   0.000     253.4364    535.6107
     _2jhedu |   57.77421   70.38791     0.82   0.412    -80.25646    195.8049
       _cons |   592.5656   204.7839     2.89   0.004     190.9845    994.1468
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```

Calculate the **F-statistic** (only appropriate for nested models!!)

$$F_{(2-1),(2295-2-1)} = \frac{\frac{0.0729-0.0726}{2-1}}{\frac{1-0.0729}{2295-2-1}} = .7416$$

https://www.danielsoper.com/statcalc/calculator.aspx?id=4

Calculate the **F-statistic** (only appropriate for nested models!!)

$$F_{(2-1),(2295-2-1)} = F_{(1),(2292)} = \frac{\frac{0.0729-0.0726}{2-1}}{\frac{1-0.0729}{2295-2-1}}$$

$$=.7416$$

F=0.7416 < **3.84** → We can not reject the null that the two R^2 are the same

- How to test if each year own education has the same effect on earnings as each years of household head’s education?

How test if each year own education has the same effect on earnings as each years of household head's education?

- Estimate (a) $Earn_i = \beta_0 + \beta_1 educ_i + \beta_2 HHeduc_i + \varepsilon_i$

and Test for H_0:$\beta_1 - \beta_2$=0 using a WALD test

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```

```
. test _2bgedu=_2jhedu
```

Wald test

```
 ( 1)  _2bgedu - _2jhedu = 0

       F(  1,  2292) =    5.94
            Prob > F =    0.0149
```

Value<0.05 ➔ The difference is significant